

Analysis and Reduction of EMI-Induced Timing Jitter in Voltage-to-Time Converters Using Differential Architectures

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Abstract—Voltage-to-Time Converters (VTCs) are key interface blocks in time-mode signal processing because they convert an analog voltage into a delay or a time difference. The same property that makes a VTC useful also makes it vulnerable: electromagnetic interference (EMI) picked up at the input, reference, supply, ground, or bias nodes is converted into delay error and timing jitter. This BTP report presents the complete semester workflow. The pre-midsemester work established the baseline current-starved VCDU/VTC operation, delay measurement, PVT/Monte Carlo sensitivity, and EMI injection methodology. The post-midsemester work investigated possible differential-delay solutions. A SET-hardened complementary differential cell, an input-conditioned differential delay cell, and a basic differential-pair-controlled delay cell were implemented and studied. These attempts showed that differential signalling alone is insufficient if the control/bias path becomes single-ended or if the differential pair leaves the saturation/small-signal region. The final direction therefore compressed the rail-to-rail control voltage into a small differential swing around a 0.9 V common-mode point. Low-amplitude common-mode EMI can then be partially rejected, whereas high-amplitude or high-frequency EMI still produces residual jitter because of finite CMRR, parasitic mismatch, and nonlinear device operation.

Index Terms—Voltage-to-Time Converter, VTC, VCDU, electromagnetic interference, timing jitter, PVT variation, Monte Carlo, differential delay cell, common-mode rejection, small-signal analysis.

I. INTRODUCTION

CMOS scaling has improved digital speed and density, but it has made high-precision voltage-domain analog design harder because of reduced voltage headroom, stronger device mismatch, and increased sensitivity to supply and substrate disturbances. Time-mode signal processing (TMSP) is attractive because information is represented by delay, phase, pulse width, or edge timing rather than by a continuously varying voltage. A Voltage-Controlled Delay Unit (VCDU) is the elementary circuit that converts a voltage into delay. Two VCDUs can be arranged as a VTC: one delay path is controlled by the signal voltage and the other by a reference voltage.

However, time-domain representation does not remove analog error. It changes its form. A small voltage disturbance near the control node or supply rail changes the current that charges or discharges an internal node. The threshold-crossing instant then shifts, so EMI appears as timing jitter. The base EMI paper by Richelli *et al.* showed that VTC

susceptibility depends not only on EMI amplitude, but also on EMI frequency and initial phase [1]. This report follows that idea and extends it through post-midsemester differential-circuit exploration.

II. PROJECT WORKFLOW AND SCOPE

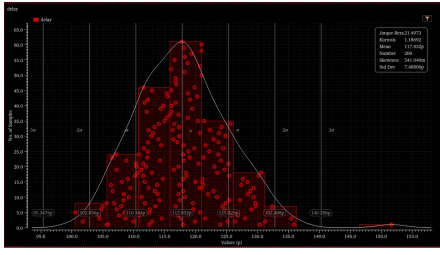
The semester workflow was not a direct jump to a final circuit. It progressed through learning, modelling, rejection of unsuitable circuits, and refinement:

- 1) Study the current-starved VCDU and its voltage-to-delay relation.
- 2) Build the baseline VTC as a two-path delay-difference circuit.
- 3) Study PVT and Monte Carlo delay spread for a single VCDU and the VTC.
- 4) Reproduce the EMI reasoning of the base paper: input, reference, supply, ground, and all-port EMI injection.
- 5) After midsemester, explore differential delay cells because common-mode EMI should cancel in a differential quantity.
- 6) Implement and test literature-inspired differential delay cells: basic differential delay cell, SET-hardened complementary differential cell, and input-conditioned differential cell.
- 7) Reject the unsuitable versions where delay was nonlinear, weakly tunable, or dominated by single-ended control paths.
- 8) Develop a compressed small-signal differential-control method so that the differential pair remains in a useful operating region.

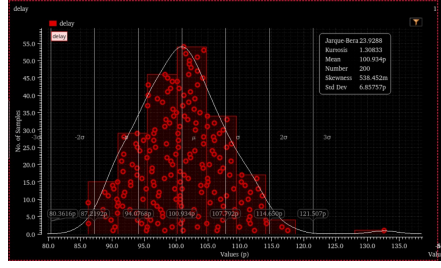
This workflow is important because it shows the project decision path. The final conclusion is not simply “use a differential pair”. The correct conclusion is that differential EMI rejection needs matched branches, valid saturation, bias symmetry, and sufficiently small differential/control swing.

III. BASELINE VCDU AND VTC OPERATION

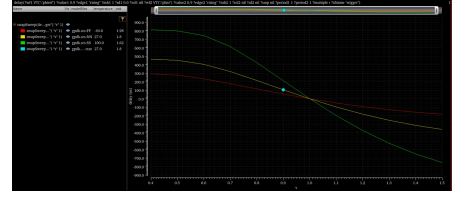
A VCDU creates delay by limiting the current available to a switching node. In a simple current-starved unit, the control voltage changes the pull-up or pull-down current. The output edge occurs when the internal capacitive node crosses the



(a) Single VCDU Monte Carlo histogram.



(b) VTC-level Monte Carlo histogram.



(c) PVT corner curves around the operating region.

Fig. 1: Pre-midsemester PVT/Monte Carlo work. A single VCDU suffers direct delay spread, while the VTC difference operation reduces common delay drift and provides a more useful operating point.

switching threshold of the next inverter. The first-order relation is

$$t_d \approx \frac{C_L \Delta V}{I_{ctrl}}, \quad (1)$$

where C_L is the effective load capacitance, ΔV is the voltage change required to reach the switching threshold, and I_{ctrl} is the current set by the control transistor. A VTC uses two such paths:

$$\Delta T_o = T(V_{in}) - T(V_{ref}). \quad (2)$$

Equation (2) is useful because common delay shifts can cancel. If both paths receive the same PVT-induced delay shift δ_{PVT} ,

$$T_1 = T(V_{in}) + \delta_{PVT}, \quad (3)$$

$$T_2 = T(V_{ref}) + \delta_{PVT}, \quad (4)$$

$$\Delta T_o = T_1 - T_2 = T(V_{in}) - T(V_{ref}). \quad (5)$$

This explains why a two-path VTC can reduce common PVT variation. But it also reveals the condition for failure: if EMI or mismatch causes unequal shifts δ_1 and δ_2 ,

$$\Delta T_{emi} = T(V_{in}) - T(V_{ref}) + (\delta_1 - \delta_2). \quad (6)$$

The residual term $(\delta_1 - \delta_2)$ is timing jitter.

IV. PRE-MIDSEMESTER PVT AND MONTE CARLO INSIGHT

Before the EMI solution was explored, delay variation had to be understood. A single VCDU has strong sensitivity to process, supply voltage, temperature, and random mismatch because all of them modify I_{ctrl} in (1). The single VCDU Monte Carlo histogram in Fig. 1 shows a mean delay near 117.8 ps with a standard deviation of about 7.49 ps. The VTC-level statistical result shows a mean around 100.9 ps with a standard deviation near 6.86 ps. The most important observation is the PVT-corner plot: the corner curves approach a common operating region, showing that using a reference path can reduce common-mode PVT drift near the chosen bias point.

V. DELAY MODEL AND EMI-TO-JITTER DERIVATION

Let $V_C(t)$ be the internal node that is discharged during a transition. The capacitor equation is

$$I_{dis} = -C_L \frac{dV_C}{dt}. \quad (7)$$

The output switches when V_C reaches V_M , approximately $V_{DD}/2$. If current is constant,

$$t_d = \frac{C_L(V_{DD} - V_M)}{I_{dis}}. \quad (8)$$

For an NMOS current-control device,

$$I_{sat} = K_n \left(\frac{W}{L} \right) (V_{GS} - V_{tn})^{\alpha_n}. \quad (9)$$

Thus, in the saturation-only approximation,

$$t_d = A(V_{GS} - V_{tn})^{-\alpha_n}, \quad (10)$$

where $A = C_L(V_{DD} - V_M)/[K_n(W/L)]$. Differentiating (10),

$$\frac{\partial t_d}{\partial V_{GS}} = -\alpha_n \frac{t_d}{V_{GS} - V_{tn}}, \quad (11)$$

$$\frac{\partial t_d}{\partial V_{tn}} = +\alpha_n \frac{t_d}{V_{GS} - V_{tn}}. \quad (12)$$

Equations (11) and (12) give two physical insights: low-overdrive operation is both nonlinear and sensitive; threshold variation directly becomes timing skew. For EMI at the control node,

$$V_{GS}(t) = V_{GS,0} + v_{emi}(t), \quad (13)$$

with

$$v_{emi}(t) = A_{emi} \sin(2\pi f_{emi} t + \phi). \quad (14)$$

At the switching instant t_s ,

$$\Delta t_d \approx -\alpha_n \frac{t_d}{V_{GS} - V_{tn}} A_{emi} \sin(2\pi f_{emi} t_s + \phi). \quad (15)$$

This equation is the core mathematical explanation of the project. It explains why amplitude, frequency, and phase matter. The VTC makes a decision at a particular threshold-crossing instant; the instantaneous EMI value at that instant decides whether the edge advances or is delayed.

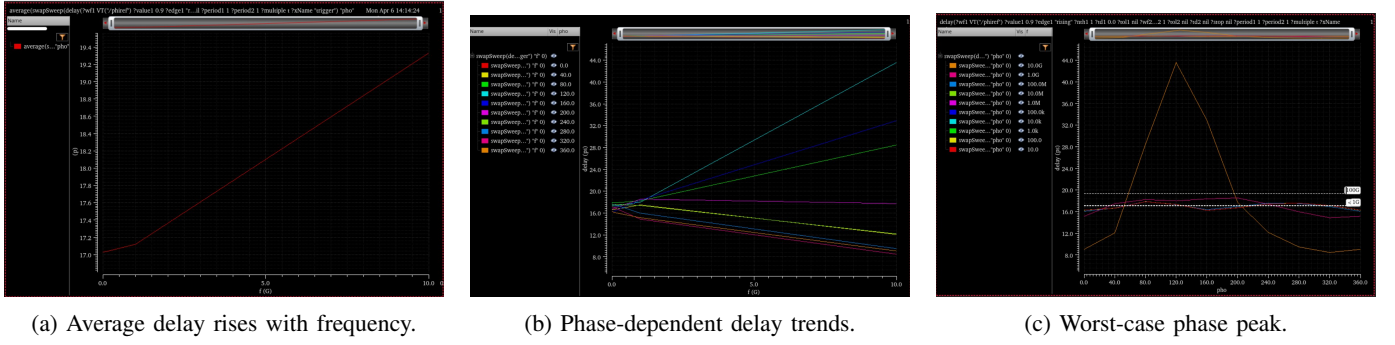


Fig. 2: Baseline VTC EMI study. Frequency increases capacitive coupling, while phase determines the instantaneous disturbance at threshold crossing.

For supply or internal-node EMI, the more general sensitivity form is

$$\Delta t_d = \frac{\partial t_d}{\partial V_{ctrl}} \Delta V_{ctrl} + \frac{\partial t_d}{\partial V_{DD}} \Delta V_{DD} + \frac{\partial t_d}{\partial V_{SS}} \Delta V_{SS}. \quad (16)$$

This is why EMI injected at different ports produces different phase/frequency trends.

VI. BASELINE VTC EMI FREQUENCY AND PHASE SWEEPS

The baseline VTC was simulated with EMI frequency and phase sweeps before proposing a new architecture. The average-delay plot in Fig. 2 shows a gradual increase with frequency. This is expected because parasitic capacitive coupling increases with dv/dt :

$$i_C = C \frac{dv}{dt} = 2\pi f_{emi} C A_{emi} \cos(2\pi f_{emi} t + \phi). \quad (17)$$

As frequency rises, the same EMI amplitude injects more displacement current into sensitive nodes.

The phase-sweep results are even more important. Some phase curves increase delay, some decrease delay, and some remain almost flat. A high-frequency worst-case phase can create a sharp delay peak. Therefore, average delay is not sufficient for EMI evaluation. A robust VTC must be evaluated using

$$\Delta t_{worst} = \max_{\phi} |\Delta t_d(\phi)|. \quad (18)$$

This pre-midsemester observation motivated the post-midsemester differential architecture.

VII. DIFFERENTIAL DELAY-CELL EXPLORATION

A differential circuit ideally rejects common-mode EMI. If

$$V_+ = V_{sig+} + v_{cm}, \quad (19)$$

$$V_- = V_{sig-} + v_{cm}, \quad (20)$$

then

$$V_+ - V_- = V_{sig+} - V_{sig-}. \quad (21)$$

The post-midsemester work studied three delay-cell ideas from literature and Cadence implementation: a basic differential delay cell, a SET-hardened complementary differential cell,

and an input-conditioned delay cell. These were not treated as final circuits; they were used to understand which properties are required for an EMI-robust VTC.

A. SET-Hardened Complementary Differential Cell

The SET-hardened complementary differential cell has cross-coupled feedback, resistive branches, and differential outputs. It is robust for fast regenerative switching, but a VTC requires smooth delay variation with input voltage. In simulation, the delay tuning was weak and nonlinear; the latch-like action improved edge regeneration but reduced controllable analog delay behaviour. Hence it was rejected as a final VTC core.

B. Input-Conditioned Differential Delay Cell

The input-conditioned cell adds auxiliary/control paths and differential outputs. It was useful conceptually because it shows that delay can be changed using both main and auxiliary paths. However, the transients showed overshoot, limited differential swing, and crossing-level sensitivity. A delay cell whose output edge is not clean cannot be used reliably as a VTC because delay extraction depends on the selected threshold.

VIII. REJECTED DIFFERENTIAL-AMPLIFIER-CONTROLLED ARCHITECTURE

A major attempted architecture is shown in Fig. 3. A differential amplifier/front-end was used, but its single-ended output node was directly connected to the tail-current control of a differential CMOS delay/inverter block. This was a natural first idea, but it had a fundamental weakness: the signal became single-ended before controlling the delay current. Therefore, common-mode EMI and supply noise could still enter the control node.

In small-signal form, a differential pair gives

$$i_+ = \frac{I_{tail}}{2} + \frac{g_m v_d}{2}, \quad (22)$$

$$i_- = \frac{I_{tail}}{2} - \frac{g_m v_d}{2}. \quad (23)$$

If only one output node is used as a control voltage,

$$V_b = A_d v_d + A_{cm} v_{cm} + A_{ps} v_{dd}. \quad (24)$$

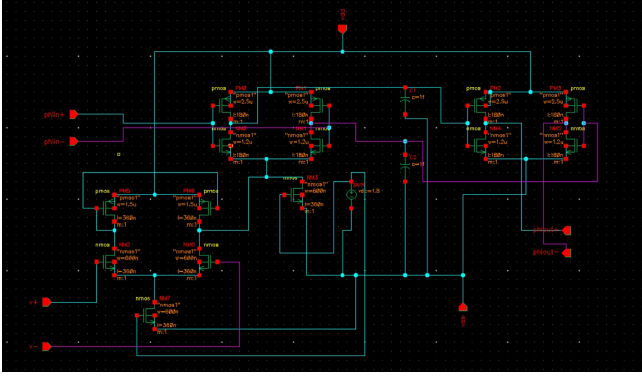


Fig. 3: Rejected differential-amplifier-controlled delay architecture. The single-ended control node was directly used to control the delay-cell tail current, which breaks the full differential EMI-cancellation path.

The terms $A_{cm}v_{cm}$ and $A_{ps}v_{dd}$ remain. The starve current becomes

$$I_{ctrl} = I_0 + g_{m,tail}V_b, \quad (25)$$

so the delay becomes

$$t_d = \frac{C_L \Delta V}{I_0 + g_{m,tail}V_b}. \quad (26)$$

Thus, the single-ended node reintroduces EMI sensitivity.

IX. EMI SWEEPS ON THE INTERMEDIATE ARCHITECTURE

EMI was injected through different ports of the intermediate differential-delay architecture. The sweeps varied EMI amplitude, phase, and control voltage. The observed curves had peaks, dips, and strongly phase-dependent delay. This confirmed the mathematical model: the delay is controlled by the instantaneous EMI value at edge crossing. However, it also showed that the intermediate circuit did not provide robust common-mode cancellation. Rising and falling delays were not balanced, and the control path remained EMI-sensitive.

The key learning was that a circuit can be visually differential but functionally single-ended if the bias or control node is not symmetric. In such a case, EMI appears as current modulation and then as delay jitter.

X. CORRECTED IDEA: COMPRESSED DIFFERENTIAL CONTROL

The next issue was the rail-to-rail control sweep. A MOS differential pair behaves as a linear differential amplifier only around a suitable bias point. If the input is swept from 0 V to 1.8 V, one branch may turn off and the other may take almost the entire tail current. Then the small-signal relation no longer holds and common-mode rejection collapses.

The corrected idea was to map the full external control variable $A \in [0, 1.8]$ into a small differential signal centered at 0.9 V:

$$V_+ = 0.9 + k(A - 0.9), \quad (27)$$

$$V_- = 0.9 - k(A - 0.9), \quad (28)$$

with $k \approx 0.15$. Therefore,

$$v_d = V_+ - V_- = 2k(A - 0.9). \quad (29)$$

For $A = 0$, the inputs are approximately 0.765 V and 1.035 V; for $A = 1.8$, they swap. The pair therefore remains near a controlled common-mode point instead of being forced into rail-to-rail overdrive.

The small-signal chain becomes

$$i_o = g_m v_d, \quad (30)$$

$$V_b = A_v v_d, \quad (31)$$

$$\Delta I = g_{m,tail} V_b = g_{m,tail} A_v v_d. \quad (32)$$

The delay is

$$t_d = \frac{C_L \Delta V}{I_0 + \Delta I}. \quad (33)$$

For $|\Delta I| \ll I_0$,

$$\Delta t_d \approx -\frac{t_{d0}}{I_0} \Delta I = -\frac{t_{d0}}{I_0} g_{m,tail} A_v v_d. \quad (34)$$

The sign depends on whether increasing current advances or delays the selected edge. Equation (34) gives the useful design rule: delay control is approximately linear only when the differential signal is small and the bias current I_0 is maintained.

XI. EMI BEHAVIOUR OF COMPRESSED DIFFERENTIAL SCHEME

With compressed differential control, low-amplitude EMI that appears equally on both branches is partly rejected. The ideal relation is

$$V_+ = V_{CM} + \frac{v_d}{2} + v_{cm}, \quad (35)$$

$$V_- = V_{CM} - \frac{v_d}{2} + v_{cm}, \quad (36)$$

$$V_+ - V_- = v_d. \quad (37)$$

In timing form,

$$T_+ = T(V_+) + \delta_{cm}, \quad (38)$$

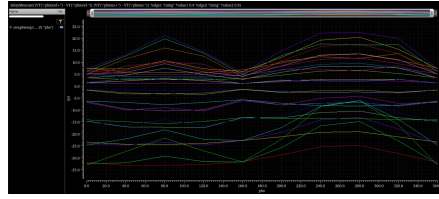
$$T_- = T(V_-) + \delta_{cm}, \quad (39)$$

$$\Delta T = T_+ - T_- = T(V_+) - T(V_-). \quad (40)$$

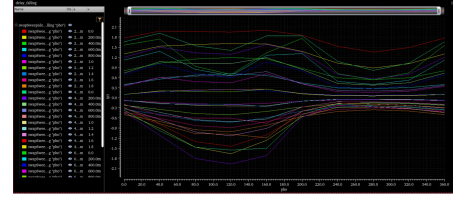
The common disturbance cancels ideally. The Cadence observations matched this partially: lower amplitude and lower frequency disturbances were more controlled, while high-amplitude and high-frequency disturbances caused residual jitter. The reason is practical rather than conceptual: at high frequency, parasitic capacitors create unequal coupling; at large amplitude, devices leave their small-signal region; with finite matching, common-mode signals convert into differential error.

Variable	Value	Device	Range Type	From	To	Step Mode	Total Steps	Inclusion List	Exclusion List
a	0.4	✓	From/To	0.2	0.6	Auto	3		
phi	150	✓	From/To	0	360	Auto	10		
v	0.9	✓	From/To	0	1.8	Auto	10		

(a) EMI sweep variables.

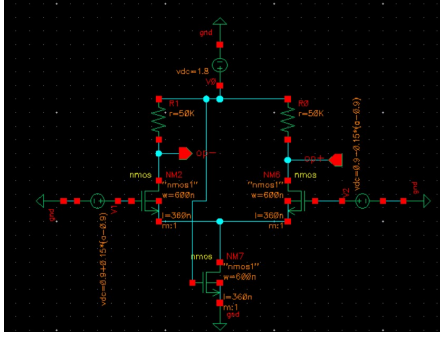


(b) Rising-delay phase dependence.

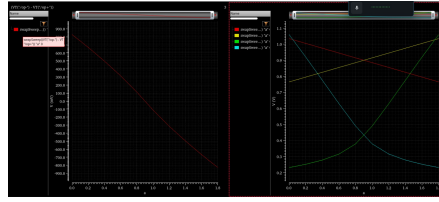


(c) Falling-delay mismatch.

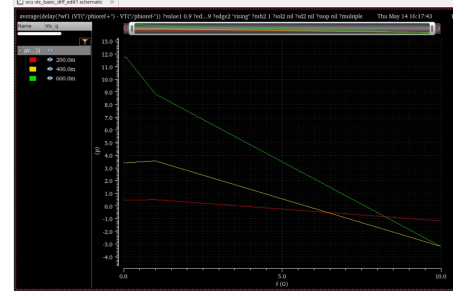
Fig. 4: EMI sweep on the intermediate architecture. The response is phase-dependent and irregular, proving that the first differential attempt did not fully reject common-mode/supply disturbances.



(a) Differential-pair DC exploration.



(b) Compressed control and output trend.



(c) Frequency/phase EMI result.

Fig. 5: Compressed differential control. The full rail-to-rail external control is converted into a small differential swing around 0.9 V; this improves common-mode rejection for low-amplitude EMI but does not remove high-frequency residual jitter.

XII. WHY THE SECOND EDITED MIRROR VERSION WAS REJECTED

A second edited version mirrored the single-ended output current of the differential amplifier to control the tail current of the differential CMOS delay cell. In the sweep, the delay stayed almost constant. This means that the mirrored current was not controlling the dominant delay mechanism. Possible reasons are: the control swing was too compressed to modulate the tail current strongly; the current mirror transistors were not in the intended saturation region; the output delay was dominated by a different path or capacitance; or the mirror was connected to a node that did not set the actual output transition current. This was an important negative result because it showed that adding a current mirror is not enough unless the mirrored current is the current that actually charges or discharges the delay node.

XIII. CHALLENGES AND LESSONS

The main technical lessons are:

- EMI in a VTC is timing error, not merely voltage ripple.
- Average delay is insufficient; worst-case phase must be checked.
- A dual VCDU/VTC reduces common PVT shifts, but EMI can remain if coupling is unequal.
- Differential circuits reject common-mode EMI only when the branches are matched and biased in the correct operating region.

- Converting a differential signal into a single-ended control node can destroy EMI cancellation.
- Direct rail-to-rail excitation of a differential pair is not valid for small-signal CMRR analysis.
- Compressed differential control around 0.9 V preserves the useful differential-pair region and gives a realistic architecture direction.

XIV. CONCLUSION

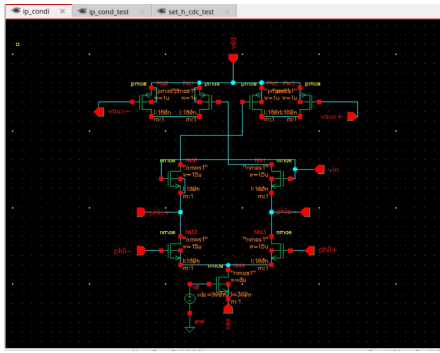
The BTP studied EMI effects in VTCs and explored differential architectures for reduction of EMI-induced timing jitter. The pre-midsemester work established the baseline VTC, PVT/Monte Carlo behaviour, and EMI phase/frequency dependence. The post-midsemester work showed why several differential delay cells were not suitable as final VTC cores: their delay was nonlinear, weakly tunable, or affected by single-ended control paths. The final useful direction was compressed differential control. It keeps the differential pair near a common-mode operating point and therefore improves common-mode EMI rejection for low-amplitude disturbances. The remaining limitation is high-frequency and high-amplitude EMI, where parasitic mismatch, finite CMRR, and nonlinear operation create residual jitter. Future work should include transistor-region tables, extracted-layout parasitics, CMRR-versus-frequency analysis, and a fully matched differential layout.

APPENDIX A

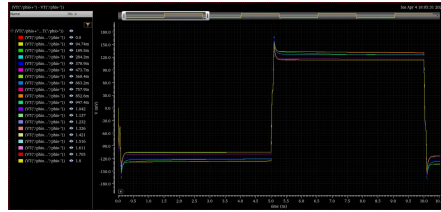
SCHEMATIC AND RESULT EVIDENCE LOG

This appendix keeps the report body readable while still documenting the important Cadence schematics and waveforms. Each figure is included with the design insight obtained from it rather than as an isolated image.

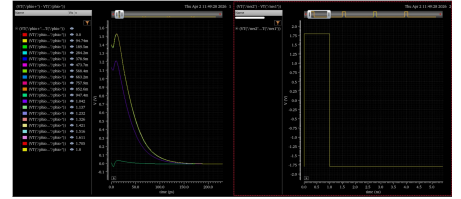
- A. SET-Hardened and Input-Conditioned Delay Cells*
- B. Basic Differential Delay Cell Experiments*
- C. EMI Injection Through Different Ports*
- D. Compressed Differential Control and Remaining Limits*



(a) SET-H CDC internal schematic.

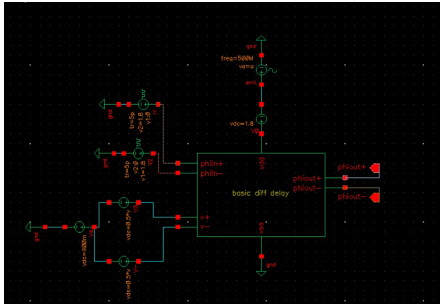


(b) SET-H CDC delay response.

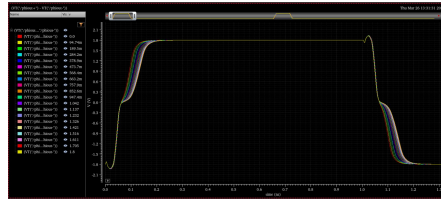


(c) Input-conditioned cell schematic.

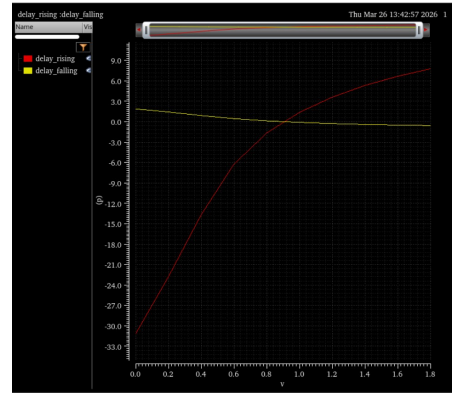
Fig. 6: Differential delay-cell exploration. The SET-H cell gave robust regenerative switching but weak analog delay tunability. The input-conditioned cell introduced auxiliary control but suffered from waveform quality and threshold-crossing sensitivity.



(a) Single differential delay-cell testbench.

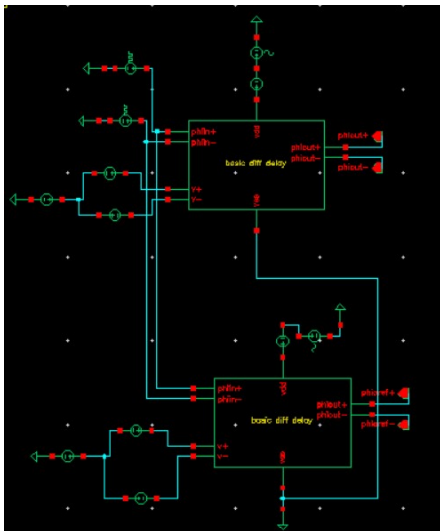


(b) Differential output transient.

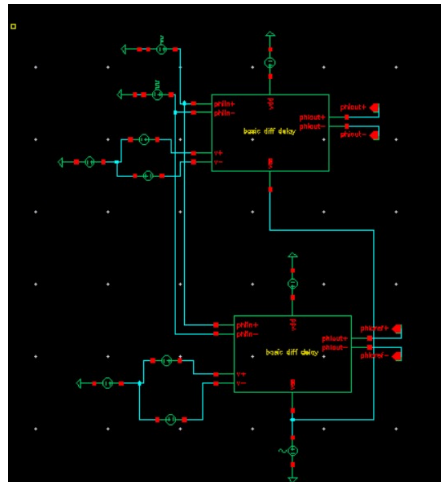


(c) Dual-cell delay measurement.

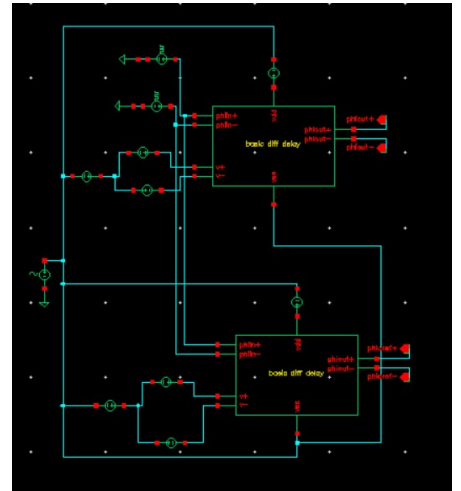
Fig. 7: Basic differential delay-cell simulations. These results confirmed that differential outputs can be generated, but rising/falling delay imbalance and nonlinear delay variation made the cell unsuitable as a precision VTC core.



(a) Differential-pair EMI setup.

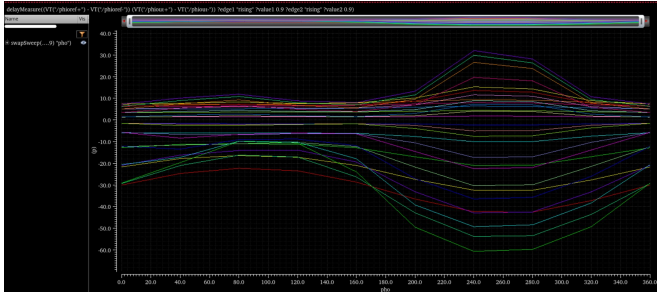


(b) Port-specific EMI setup.

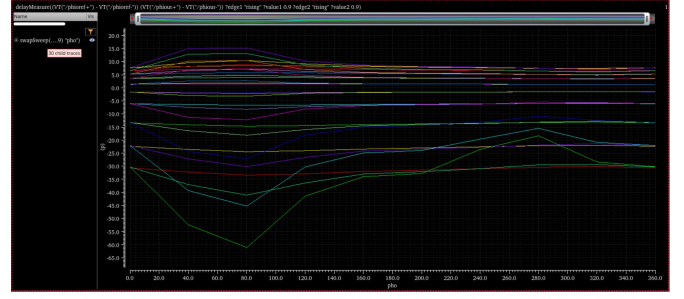


(c) All-port/common-mode EMI setup.

Fig. 8: EMI was injected through different locations to understand which nodes control timing. Supply and bias/control injection affected delay strongly because they modulated the current that charges/discharges the timing node.

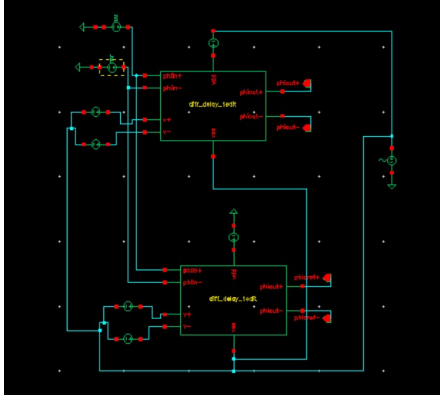


(a) Port EMI phase response.

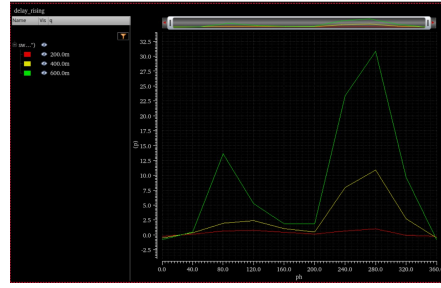


(b) All-port EMI response.

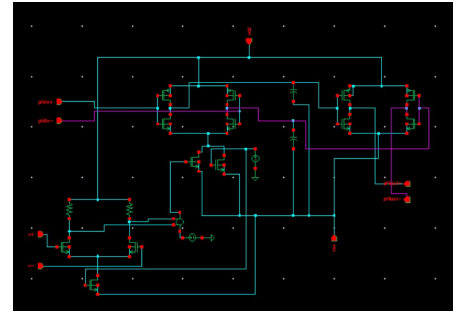
Fig. 9: Representative EMI responses. The irregular phase curves show that EMI-induced delay is not a simple DC offset; it depends on where the sinusoidal disturbance is at the switching instant.



(a) First edited compressed-control testbench.

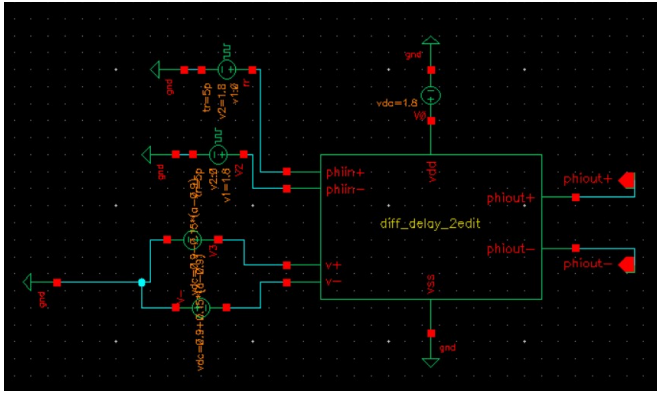


(b) Phase-dependent delay after compression.



(c) Delay trend versus frequency.

Fig. 10: Compressed differential control improved the operating-region problem but did not eliminate all EMI-induced jitter. High-frequency EMI produces unequal parasitic coupling and residual differential error.



(a) Second edited mirror-control testbench.



(b) Delay remained almost constant.

Fig. 11: Second edit rejected. The mirrored-current path did not effectively control the dominant delay current; hence the delay did not respond to the input sweep.

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